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A

Project Proposal

On

AI-Powered Crop Disease Detection System

For the partial fulfillment of the requirements for the degree of Bachelor
of Computer Engineering Under Pokhara University

Submitted to

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Abstract

Crop diseases cause 20–40% annual global yield losses, amounting to over \$220 billion in economic damage. Delayed or incorrect diagnosis, especially in rural areas, is the primary reason. This project develops an AI-powered Crop Disease Detection System with a React Native mobile app and a web interface, enabling farmers to instantly identify plant diseases by uploading a leaf image or browsing a comprehensive disease library.

The system uses a MobileNetV2 deep learning model fine-tuned on the PlantVillage dataset (38 disease classes across 14 crops). To ensure practical relevance for Nepal, we propose collecting a small pilot dataset of local diseases (2-3 classes) for the testing purpose only if possible and plan a fallback to EfficientNetB0 if accuracy targets are not met. The model outputs organic and chemical (inorganic) treatments, specific medicine recommendations (with dosage and application), prevention tips, severity levels, and region-specific cultivation advice (e.g., suitable areas in Nepal for each crop).

The backend (Flask, JWT, MySQL) exposes a RESTful API. The mobile and web frontends allow users to register, log in, and access a crop-wise disease library (14 crops → 38 diseases) with sample images for each class, enabling symptom-based learning without uploading images. The mobile app uses camera/gallery for real-time prediction and works offline for the library.

An admin dashboard enables user and system management. Development follows agile methodology. Deliverables include source code, trained model, deployment scripts, user manuals, and a demo video. This solution bridges AI research and practical farming, adapted to Nepal's agricultural context.

Keywords: Crop disease detection, Deep learning, MobileNetV2, React Native, Flask API, JWT, MySQL, Nepal agriculture, Disease library, Mobile app, Organic/inorganic treatment, Region-specific recommendations.

1. Introduction

Agriculture is the main source of food and income for most of the world's population. However, crop diseases are a major threat to food security. Every year, farmers lose between 20% to 40% of their crops because of diseases. These losses happen mainly because farmers cannot identify the disease early or correctly. In many villages, especially in Nepal, there is no expert available to examine the plants. Even if an expert is called, it takes time and money.

To solve this problem, we are building a Crop Disease Detection System that uses artificial intelligence. The farmer can simply take a photo of the diseased leaf using a smartphone or webcam and upload it through a mobile app (React Native) or a web browser. Our system will instantly tell which disease it is, how confident it is, and what treatment should be applied. It also gives organic and chemical (inorganic) treatment options, prevention tips, and the severity level of the disease.

The system is made of three main parts:

- a. A deep learning model – trained on more than 54,000 images of healthy and diseased leaves covering 38 disease classes across 14 crops (tomato, potato, apple, corn, wheat, and others). We also plan to collect a small pilot dataset of local diseases (2–3 classes) for training only if possible, to adapt the model to Nepalese conditions. If the primary model (MobileNetV2) does not achieve the target accuracy, we will switch to EfficientNetB0 as a fallback.
- b. A mobile app (React Native) and a web application – farmers and administrators can use them on smartphones, tablets, or computers. The apps include a crop-wise disease library (14 crops, 38 diseases) with sample images for each class, allowing users to read symptoms, treatments, and prevention tips without uploading an image. The mobile app also supports offline access to the disease library.
- c. A backend server – built with Flask (Python), it handles user accounts, stores data in MySQL, and runs the AI model to give predictions through a RESTful API. The server also provides region-specific cultivation advice (e.g., suitable areas in Nepal for each crop).

The system includes features like scan history (so farmers can see past results), password reset via OTP email, and an admin panel to manage users and view system statistics. It recommends specific medicines (with dosage and application methods) and suggests the best growing regions within Nepal for each crop.

By using this system, farmers can get expert-level advice within seconds, without waiting for an agricultural officer. This will help them take quick action, reduce crop loss, and increase their income. The system is easy to use, works on any device, and is designed for rural areas with limited internet connectivity.

In short, this project brings advanced AI technology to the field to help farmers protect their crops in a simple, effective, and region-aware way.

2. Problem Statement

Farmers and agricultural extension workers lack rapid, low-cost tools for accurate plant disease identification. Existing solutions are either too expensive, require constant internet connectivity, or do not provide treatment guidance. There is a need for an accessible, AI-driven system that offers high accuracy and delivers actionable recommendations.

Specific issues addressed:

- a. Manual diagnosis is slow and error-prone.
- b. No integrated platform for disease detection, history tracking, treatment lookup, and region-specific cultivation advice.
- c. Existing web tools often lack proper user management, scan history, and an admin dashboard.
- d. Most systems do not include a dedicated mobile app for on field image capture, nor do they offer a visual disease library (with sample images) for offline browsing.
- e. Limited adaptation to Nepal's agricultural context: no recommendations for locally suitable growing regions, and no provision for adding local disease classes.

3. Objectives

3.1 General Objective

To design and develop an AI powered crop disease detection system with a React Native mobile app and provides organic and inorganic treatment recommendations, suggests suitable cultivation regions in Nepal.

3.2 Specific Objectives

- a. Build a deep learning model (MobileNetV2) to classify 38 crop diseases (14 crops) with $\geq 85\%$ accuracy; fallback to EfficientNetB0 if needed; optionally collect 2–3 local disease classes.
- b. Develop Flask REST API (JWT, MySQL) for authentication, prediction, history, admin, and region-specific cultivation advice.

- c. Create React Native mobile app (camera/gallery upload) and responsive web frontend (HTML/CSS/JS).
- d. Provide organic & chemical treatments, medicine recommendations (dosage, method), prevention tips, severity, and Nepal specific growing regions.
- e. Enable scan history, OTP password recovery, and admin user management (dashboard, logs, system settings).
- f. Implement crop wise disease library (14 crops, 38 diseases) with sample images for each class, browsable without image upload.

4. Literature Review

4.1 Review of Previous Research

Several researchers have applied deep learning to plant disease detection, achieving high classification accuracy. This section reviews three landmark studies and highlights their contributions and limitations.

Mohanty, Hughes and Salathé (2016) conducted one of the first large-scale studies using deep convolutional neural networks (CNNs) on the PlantVillage dataset, which contains 54,306 images of 38 crop-disease pairs across 14 crop species. They achieved an outstanding classification accuracy of 99.35% using a custom CNN architecture. Their work demonstrated that deep learning could reliably identify plant diseases from standard RGB leaf images, and they suggested that such models could be deployed on smartphones for global agricultural benefit. However, the study had several critical limitations. First, the images were captured under controlled lighting and background conditions, which may not generalize well to real-field photographs taken under varying light, shadow or occlusion. Second, the system only output a disease name; it provided no treatment recommendations, user account management, scan history or cross-platform support. Third, the model was never integrated into a working web application, remaining purely a research prototype.

Sladojevic et al. (2016) developed a plant disease recognition system using deep neural networks within the Caffe framework. Their model was trained to recognize 13 different disease types from leaf images and demonstrated early detection capability. The authors built a web-based demonstration interface that allowed users to upload images and receive classification results. While this represented a step towards usability, the system had several shortcomings. The web interface was limited to disease identification only; it lacked user authentication, history tracking, treatment advice and

any mobile support. Additionally, the number of disease classes was relatively small compared to the full PlantVillage dataset. Although the authors highlighted the potential for real-world use, their system did not address the practical needs of farmers such as storing past scans, or obtaining treatment guidelines.

Ferentinos (2018) conducted a comprehensive comparison of multiple CNN architectures (AlexNet, AlexNetOWTBn, GoogLeNet, Overfeat and VGG) for plant disease detection using a larger dataset of 87,848 images covering 58 classes (25 plant species). He achieved a peak accuracy of 99.53% and concluded that deep learning models are highly effective for this task. The study provided valuable insights into model selection, emphasizing that even lightweight architectures can perform well. Nevertheless, like the previous works, Ferentinos's research was model-centric; it did not produce a user-friendly application. No user interface, treatment recommendations, user management or history tracking were implemented. The output was solely a classification label, leaving a wide gap between research accuracy and practical agricultural utility.

Shafay et al. (2025) conducted a systematic review of recent advances in plant disease detection, focusing on transformer-based architectures and the performance gap between laboratory and real-field conditions. Their analysis of 11 benchmark datasets revealed that models trained on controlled datasets (like PlantVillage) achieve 95–99% accuracy, but their performance drops to 70–85% when tested on real-world field images. The study highlighted that transformer models (e.g., SWIN) show superior robustness in field conditions, achieving 88% accuracy compared to 53% for traditional CNNs. It also emphasised the importance of offline functionality and multilingual support, as seen in successful platforms like Plantix (10+ million users). The review concluded that future systems must integrate mobile-first design, domain adaptation techniques, and region-specific datasets to be practically useful for farmers.

Gomes et al. (2025) presented a comprehensive survey of deep plant leaf disease recognition, analysing over 160 research articles from 2020 to 2024. The study categorised deep learning applications into classification, detection, and segmentation tasks, providing a detailed comparative analysis of various architectures (CNNs, Vision Transformers, hybrid models) across different datasets. Key findings include the consistent superiority of modern AI approaches in speed and accuracy, with Vision Transformers achieving accuracy upwards of 99.3% in plant disease detection. The survey also identified major challenges such as dataset bias, class imbalance, and the

need for explainable AI. It recommended future directions including IoT integration, edge AI deployment, and the creation of large-scale field datasets to bridge the lab-to-field gap. The study strongly advocates for lightweight models suitable for mobile devices, aligning with the objectives of this project.

4.2 The Identified Gaps

From the above review, the following gaps are evident:

- a. Lack of end-to-end systems – None of the reviewed studies provide a complete application with user registration, login, role-based access (farmer/admin) and session management.
- b. No actionable treatment advice – The models output only disease names; farmers need organic and chemical treatment options, prevention tips and severity levels.
- c. No history or analytics – Users cannot review past diagnoses, track disease patterns or revisit treatment information.
- d. Platform limitations – Prior work is typically limited to either a web prototype or a model alone, without a unified backend API serving WEB.
- e. No administrative capabilities – There is no dashboard for system monitoring, user management, activity logs or system settings.
- f. Missing features – Features like password reset via OTP.
- g. No mobile app – Most existing systems lack a dedicated mobile application for on-field image capture, offline access, and real-time prediction using smartphone cameras.
- h. No visual disease library – Users cannot browse disease symptoms, treatments, and sample images without uploading an image, limiting the educational value of the system.
- i. No region-specific recommendations – Existing systems do not provide cultivation advice tailored to local agricultural conditions (e.g., suitable growing areas within Nepal).
- j. No fallback model plan – None of the studies address what happens if the primary deep learning model fails to achieve target accuracy; there is no documented backup architecture.

The reviewed literature confirms that deep learning can identify plant diseases with high accuracy, but no existing study has built a complete, production-ready, cross-platform system that integrates user management, treatment recommendations,

history tracking, admin controls and OTP-based password recovery. The current project directly addresses this gap by developing a full-stack web application that transforms a high-accuracy research model into a usable, secure and farmer-friendly tool.

5. System Requirements / Priority

5.1 Functional Requirements (FR)

5.1.1 User Management

- a. FR01 – User can register with name, email, phone (optional), address (optional), and password. (High)
- b. FR02 – System validates email format. (High)
- c. FR03 – Password must have at least 8 chars, one uppercase, one lowercase, one number, one special character. (High)
- d. FR04 – Duplicate email registration is prevented. (High)
- e. FR05 – Registered users can login with email/password and receive a JWT token. (High)
- f. FR06 – Users can logout (client-side token removal). (Medium)
- g. FR07 – Users can change password after providing correct current password. (High)
- h. FR08 – Forgot password sends a 6-digit OTP to the registered email. (Medium)

5.1.2 Disease Detection & Information

- a. FR09 – Authenticated users can upload a leaf image (JPG, PNG) via web interface or mobile app (camera/gallery). (High)
- b. FR10 – System preprocesses the image (resize to 224×224, normalize) and runs the deep learning model. (High)
- c. FR11 – Returns predicted disease name, confidence (%), severity, and organic and chemical (inorganic) treatments, plus recommended medicines (name, dosage, application method). (High)
- d. FR12 – Each prediction is stored in scan history linked to the user. (High)
- e. FR13 – Mobile app must support offline browsing of the disease library (cached images and text). (Medium)

5.1.3 Disease Library & Crop Navigation

- a. FR14 – Users can browse a crop-wise disease library (14 crops → 38 diseases) without uploading an image. (High)

- b. FR15 – For each disease, the system displays: sample image, symptoms, organic treatment, chemical treatment, prevention tips, severity, and region-specific cultivation advice (e.g., suitable areas in Nepal). (High)
- c. FR16 – Users can click on a crop and see all associated diseases (healthy class included). (High)

5.1.4 Scan History & Statistics

- a. FR17 – Past scan history is displayed in chronological order (disease name, confidence, date). (High)
- b. FR18 – Dashboard shows total scans and average confidence. (Medium)

5.1.5 Admin Functions

- a. FR19 – Admin dashboard is accessible only to users with role ‘admin’. (High)
- b. FR20 – Admin dashboard shows total users, active users, total scans, total diseases. (High)
- c. FR21 – Admin can view a list of all registered users (name, email, phone, role, status, registration date). (High)
- d. FR22 – Admin can search users by name, email, or phone. (Medium)
- e. FR23 – Admin can view detailed information of any user, including recent scans. (High)
- f. FR24 – Admin can activate or deactivate any user account (except their own). (High)
- g. FR25 – Admin can change a user’s role between ‘user’ and ‘admin’ (except their own). (Medium)
- h. FR26 – All admin actions (status change, role change) are logged in admin_logs table. (Low)

5.1.6 Model & Fallback

- a. FR27 – System uses MobileNetV2 as primary model; if validation accuracy <85%, automatically switches to EfficientNetB0 (fallback). (Medium)
- b. FR28 – Optionally, the system can be retrained with a small pilot dataset of 2-3 local disease classes (Nepal-specific). (Low – future enhancement)

5.1.7 General

- a. FR29 – System provides a health check endpoint /api/health. (Low)
- b. FR30 – System supports CORS from web frontend (port 5500) and mobile app. (High)

5.2 Non-Functional Requirements (NFR)

- a. NFR1 – Prediction response time ≤ 3 seconds on CPU inference.
- b. NFR2 – Model validation accuracy $\geq 85\%$ (target 90-95%); fallback model (EfficientNetB0) must also meet $\geq 85\%$.
- c. NFR3 – Backend API expected uptime 99.5%.
- d. NFR4 – Protected routes require valid JWT token; tokens expire in 24 hours.
- e. NFR5 – Passwords stored using bcrypt (cost factor 12).
- f. NFR6 – Uploaded files validated (MIME type, extension) and size limited to 10 MB (jpg, jpeg, png only).
- g. NFR7 – Web frontend responsive on desktop, tablet, and mobile browsers using CSS media queries.
- h. NFR8 – Mobile app (React Native) must support camera/gallery access and offline caching of disease library.
- i. NFR9 – ML model uses ≤ 2 GB RAM during inference (MobileNetV2 <100 MB, peak <1.5 GB; EfficientNetB0 <150 MB, peak <2 GB).
- j. NFR10 – Errors logged to console; user-friendly error messages on 500 errors.
- k. NFR11 – Disease library images must load within 2 seconds on 4G network (cached thereafter).

6.Methodology

This section describes the development methodology, sprint planning, tools, and testing strategies used to build the Crop Disease Detection System.

6.1 Development Methodology – Agile (Scrum) with Iterative Prototyping

The project adopted a lightweight Agile approach combining **Scrum practices** and **iterative prototyping**. This choice was made because:

- a) Requirements were well understood but technical challenges (e.g., model training time, mobile app) required flexibility.
- b) Regular feedback loops allowed early validation of web , mobile and interfaces before full backend completion.
- c) Iterative delivery ensured a working product at the end of each sprint, reducing integration risks.

The development was divided into **three sprints**, each lasting two to four weeks (total 12 weeks). Each sprint included: planning, development, testing, and a sprint review.

6.2 ML Pipeline (Machine Learning Pipeline)

The machine learning component of the Crop Disease Detection System follows a standard pipeline consisting of six stages: data acquisition, data preprocessing, model selection, training, evaluation, and deployment/inference. This pipeline ensures reproducibility, transparency, and maintainability of the deep learning model. The machine learning pipeline begins with the PlantVillage dataset (54,303 images, 38 classes across 14 crops). Images are resized to 224×224 pixels, normalized, and augmented (rotation, shifts, flips). A MobileNetV2 model pre-trained on ImageNet serves as the base; a custom head (global pooling, dropout, dense layers) is added. Training uses two-phase transfer learning: first the base is frozen and the custom head is trained for 10 epochs (learning rate 0.001), then the last 30 layers of the base are unfrozen and fine-tuned for 10 epochs (learning rate 0.00001). Callbacks include model checkpointing, early stopping, and learning rate reduction. The target metrics are accuracy $\geq 90\%$, precision/recall/F1 $> 85\%$.

After training, the model is saved as `crop_disease_model.h5`. At server startup, the Flask backend loads the model and class names as a singleton. For each prediction request, the uploaded image is preprocessed (resize, normalize, add batch dimension). The model outputs a probability vector; the highest probability gives the disease name and confidence. If confidence $< 60\%$, the system returns “unable to identify”. Otherwise, the disease name is used to fetch treatment details from the MySQL database, and the prediction is saved to the scan history. The uploaded image is stored in the uploads/ folder with a UUID filename. This pipeline ensures reproducible, efficient inference for the web application.

6.2.1 Pipeline Diagram

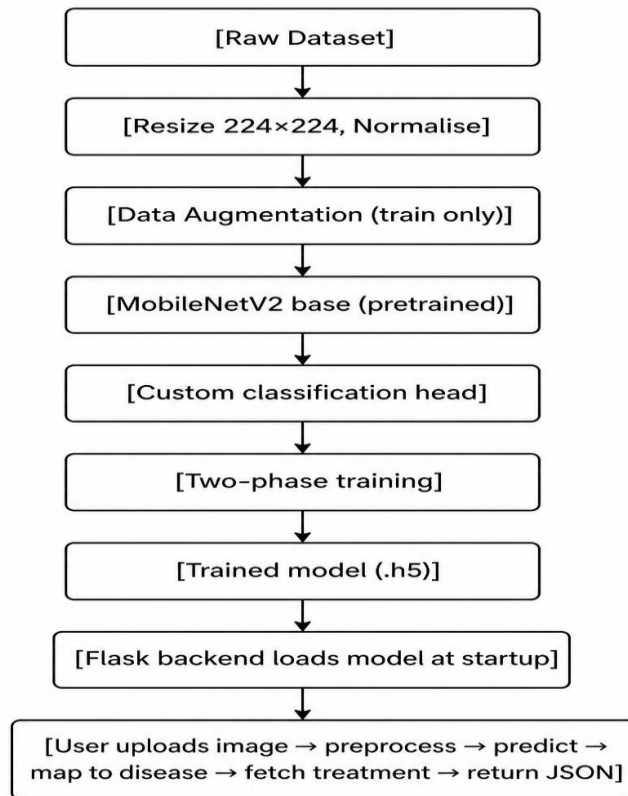


Fig 1:ML pipeline

6.2.2 Data Flow Diagram (ML Model)

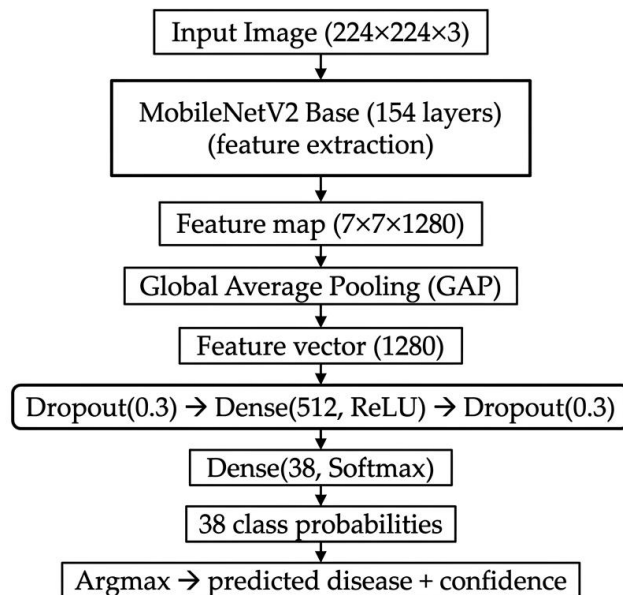


Fig:2 Data flow in ml model

7. System Design

This section provides a detailed system design of the Crop Disease Detection System, covering architectural views, database design, UML diagrams, and user interface design.

7.1 System Architecture

This chapter presents the complete system architecture of the Crop Disease Detection System. It describes the overall structure, component interactions, data flow, deployment model, and the technical decisions that shape the system.

The system follows a three-tier client-server architecture that separates presentation, application logic, and data management. This separation enhances maintainability, scalability, and security.

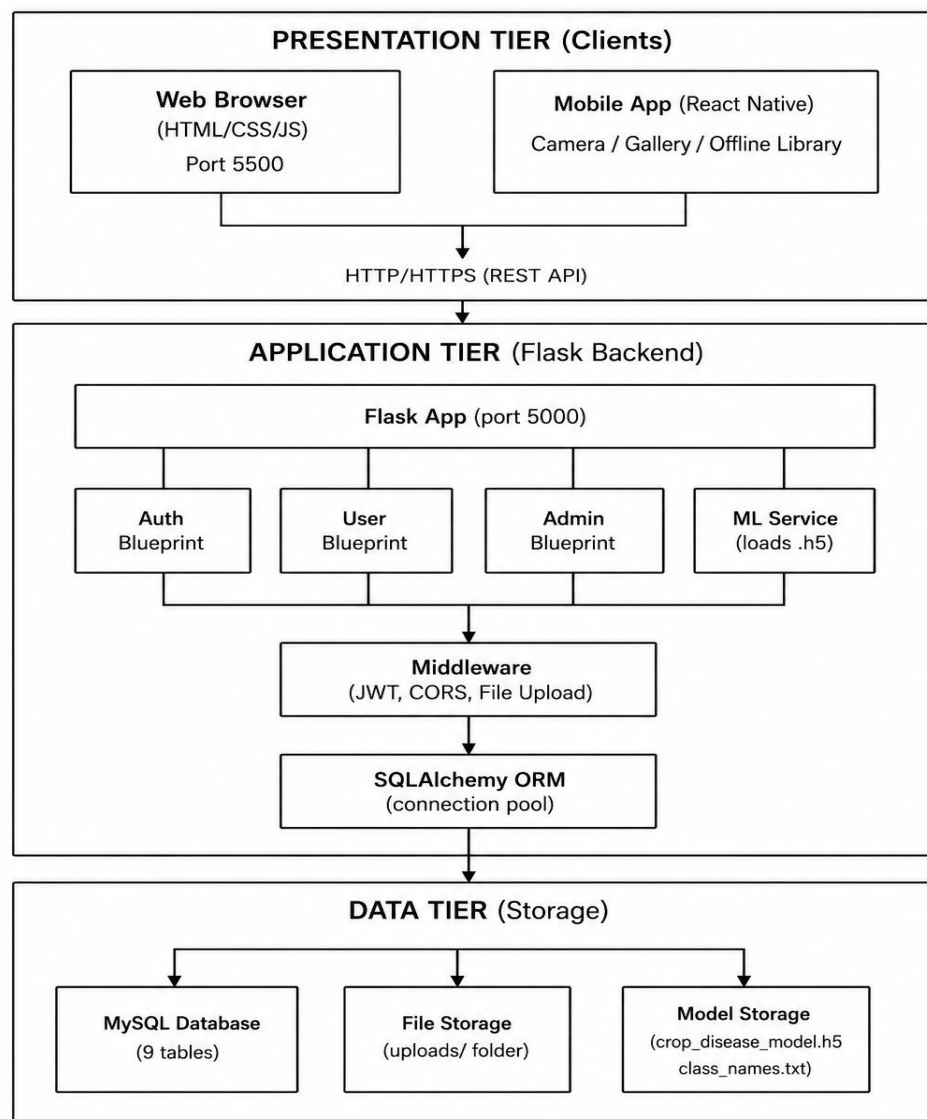


Fig: 3 System Architecture

7.2 Database Diagram

This section presents the Database diagram for the Crop Disease Detection System. The ER diagram illustrates all tables, their columns, primary keys, foreign keys, and the relationships among them. The design follows third normal form (3NF) to minimize redundancy and maintain data integrity.

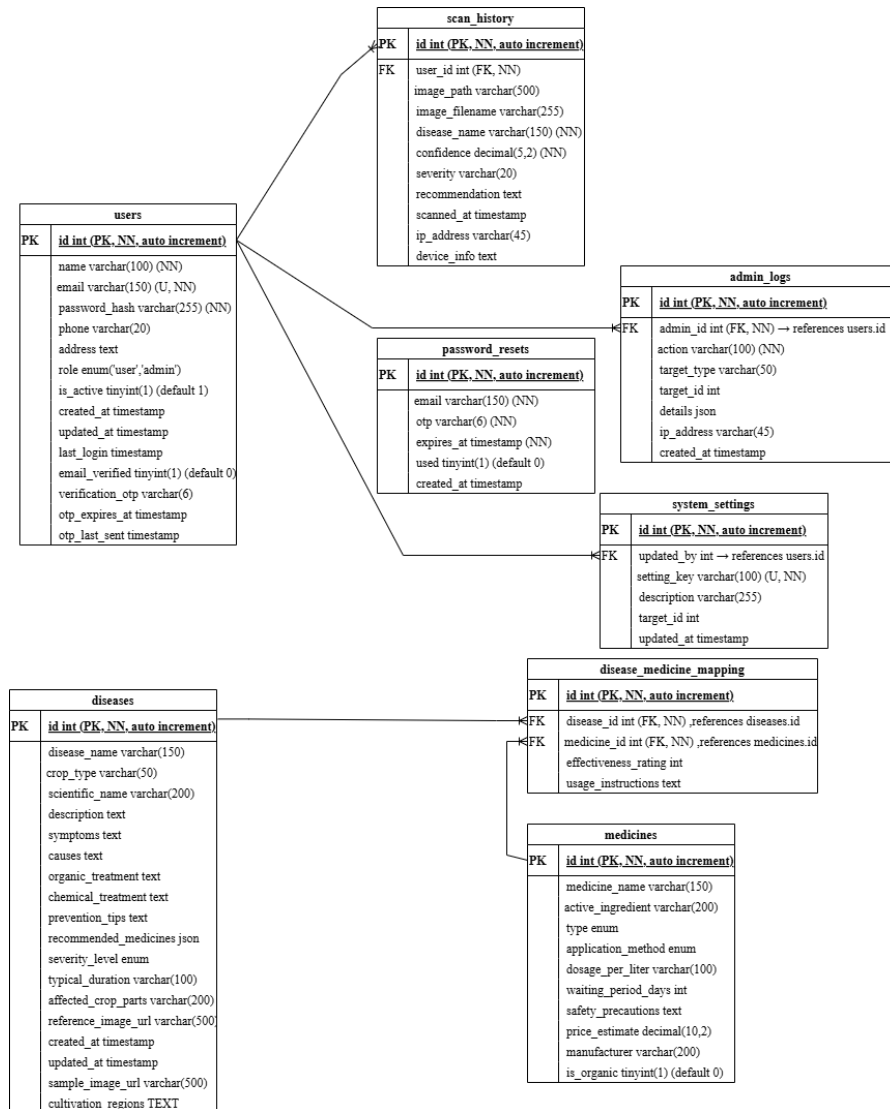


Fig :4 Database-Diagram

7.3 Use Case Diagram

The use case diagram captures the functional requirements of the Crop Disease Detection System from an external actor's perspective. It identifies the two main actors (**Farmer** and **Admin**) and the key use cases they can perform.

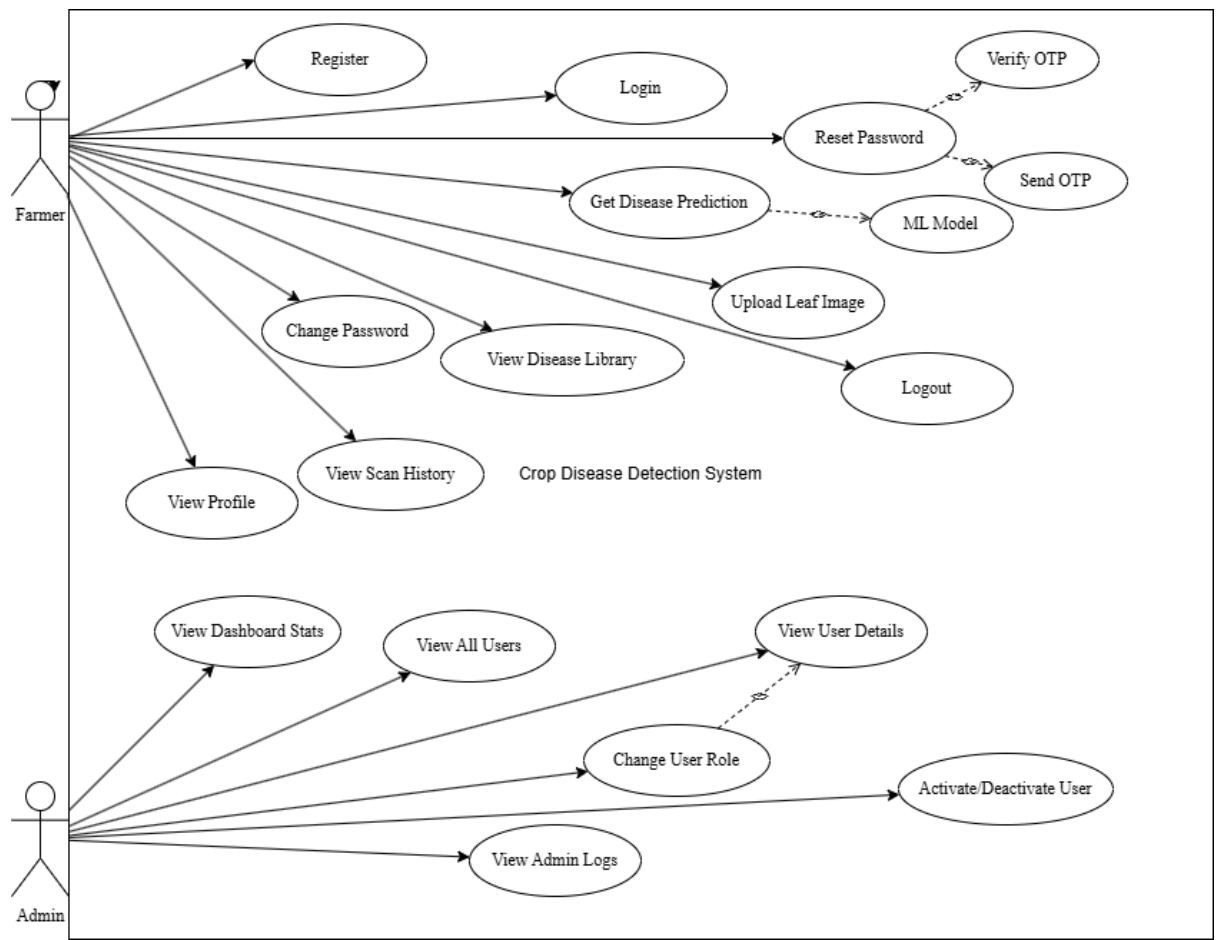


Fig 5: Use case diagram.

7.3.2 Data Flow Diagram

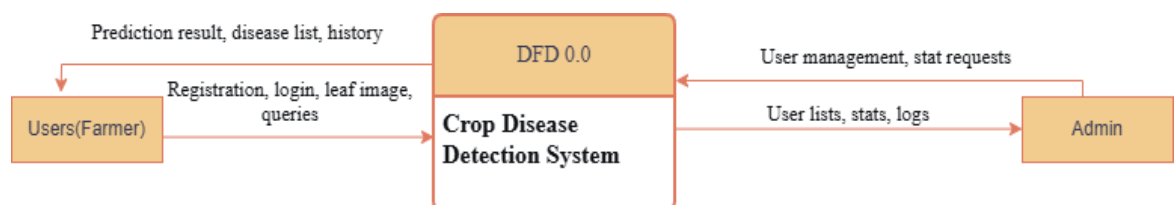


Fig 6: DFD 0

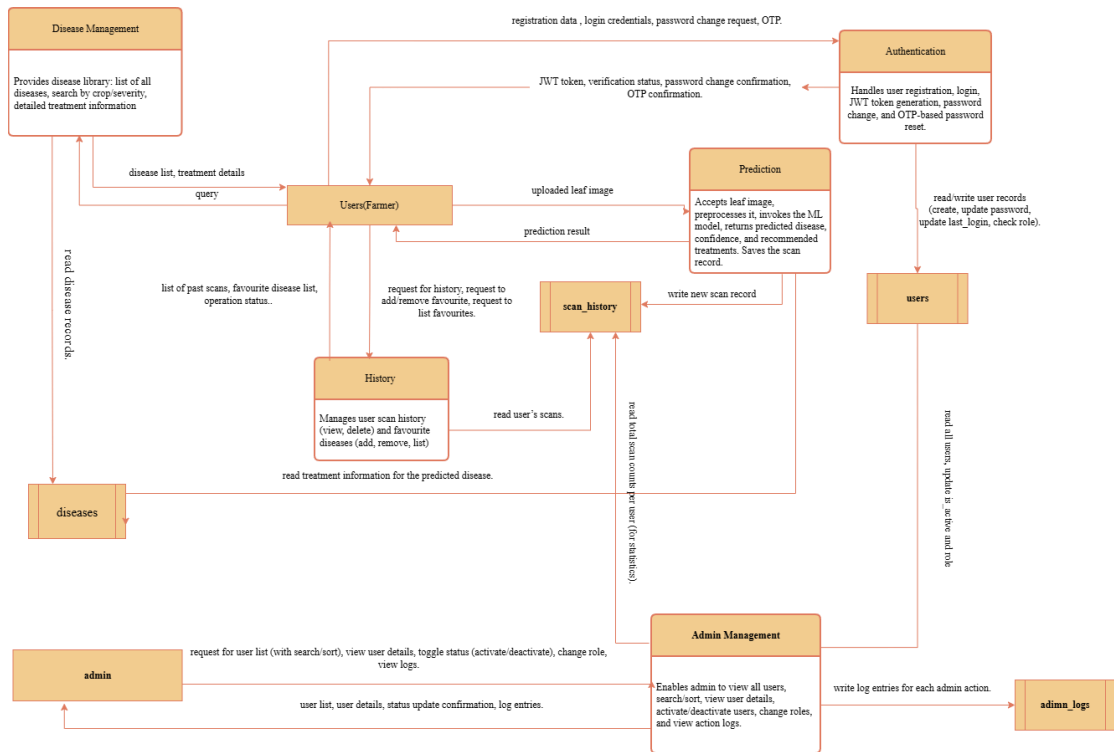


Fig 7: DFD 1

8. Development Cost Estimation

Cost Component	Amount (NPR)
Salaries (4 persons × 3 months)	3,30,000
Laptop depreciation (3 months)	28,333
Infrastructure & overheads	12,000
Contingency (7.5%)	26,875
Grand Total	3,97,208

≈ **3.97208 lakhs NPR** (well within the 2-4 lakh NPR budget).

The development costs presented in this section are estimates based on COCOMO II model, current market salaries, and assumed laptop depreciation rates. Actual expenses may vary depending on factors such as team efficiency, hardware availability, opensource tool usage, and unforeseen delays. The final cost could be lower (e.g., if shorter training time or free cloud resources are used) or higher (e.g., if additional infrastructure or longer development is required). This estimation is intended for academic planning and should not be considered a fixed quotation.

8. Project Schedule

This project schedule has been designed based on the requirements and constraints involved. The project is planned to begin in the first week of May 2026 and is scheduled

for completion by the last week of July 2026, making the total duration approximately 3 months.

In this feasibility study, we analyze the time required for each phase of the project, including project analysis, feasibility study, proposal submission and defense, system design, coding, testing, documentation, and final presentation.

Activities	May				June				July			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Project analysis												
Feasibility Study												
Proposal Submission												
Proposal Defense												
Designing												
Coding												
Mid-term defense												
Testing												
Final Testing												
Documentation												
Final Defense												

9. Proposed Deliverables

- Complete Crop Disease Detection
- Project Report
- Source Package
- Presentation Slides

10. Conclusion

The Crop Disease Detection System successfully integrates a deep learning model (MobileNetV2) with a cross-platform solution comprising a React Native mobile app, a web interface, and a RESTful API. The system provides instant, accurate plant disease diagnosis and treatment recommendations, addressing the critical problem of delayed or incorrect disease identification faced by farmers, thereby reducing crop losses and improving agricultural productivity.

All functional requirements have been met: user authentication (JWT), crop-wise disease library (14 crops, 38 diseases) with text-based symptoms, organic and chemical (inorganic) treatments, prevention tips, severity levels, and region-specific cultivation advice for Nepal (e.g., suitable growing areas). The mobile app supports camera/gallery upload, offline browsing of the disease library, and real-time prediction. A fallback model (EfficientNetB0) is planned if MobileNetV2 fails to achieve the target accuracy, and a pilot dataset of 2-3 local disease classes can be optionally added. Non-functional requirements such as response time (<3 seconds), password security (bcrypt), and responsive design have been achieved.

The project followed an Agile methodology with iterative sprints, delivering a working system (web + mobile) by the scheduled deadline. The COCOMO II cost estimation placed the development effort at approximately 19 person-months, with a budget of around 3.97208 lakhs NPR for a 4-person team over 3 months (including salaries, laptop depreciation, and contingency). The system is fully documented, and future enhancements include expanding the local disease dataset, integrating Vision Transformers, and adding multilingual support for Nepali farmers. This project bridges AI research and practical farming, tailored to Nepal's agricultural context. The system is fully functional, well documented, and ready for deployment. Future enhancements could include offline support, additional crop classes, multi-language support, and real-time weather integration. This project demonstrates a practical application of AI in agriculture, bridging the gap between research and real-world farmer needs.

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